

On the effect of surface roughness in point-contact EHL: Formulas for film thickness and asperity load

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ABSTRACT

Formulas are derived for predicting the central film thickness, minimum film thickness, and the asperity load ratio in point-contact EHL of rough surfaces. The rough EHL model includes simultaneous solution to the modified Reynolds and surface deformation equations with the consideration of both bulk and asperity deformations. To include the effect of surface roughness, a statistically-based elasto-plastic asperity micro-contact model is utilized. Regression analyses based on the results from an extensive set of simulations are performed to obtain predictive expressions for the film thickness and the asperity load ratio. These formulas are of the form $f(\kappa, W, U, G, \bar{\sigma}, V)$, where the parameters represented are ellipticity, dimensionless load, speed, material, surface roughness and hardness, respectively. The results predicted using these formulas are in good agreement with wide range of data available in the literature.

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1. Introduction

Elastohydrodynamic lubrication (EHL) is the governing lubrication regime in many industrial applications. While line-contact EHL exists in gear teeth, heavily-loaded journal bearings, cam-followers, and the like, point-contact EHL applies in components with spherical/elliptical elements such as ball bearings and artificial body joints.

The first systematic solution of point-contact EHL was proposed by Hamrock and Dowson [1,2] where they numerically solved the problem by the relaxation method. They later proposed curve-fit expressions for the central and minimum film thickness that are widely used [3]. The numerical solution methodology was subsequently improved by other researchers to include heavier loads [4,5]. Among many other contributions that followed, Venner [6] developed a fast solution method for the spherical-contact EHL using the multilevel technique which was later extended to elliptical contact [7,8]. The effect of transient condition was included in the solution by Jalali-Vahid et al. [9]. The non-Newtonian behavior of the lubricant in point-contact EHL was investigated in some recent studies [10–14].

Within the context of mixed point-contact EHL, Zhu and Cheng [15] first introduced a model based on statistical consideration of the surface asperities. For this purpose, they utilized the modified Reynolds equation derived by Patir and Cheng [16] together with

the elastic deformation of the surface asperities using the Greenwood–Tripp model [17]. Later, Venner [6] integrated the actual surface profiles into his smooth point-contact model to study the effect of roughness. Other researchers used similar methods to include the effect of actual surface profile in point-contact EHL [18,19]. A comprehensive model based on deterministic treatment of the surface roughness was first introduced by Chang [20] for line-contact EHL, and later extended to point-contact EHL [21–27]. Most recently, following the similar studies for line-contact EHL [28,29], Sojoudi and Khonsari [30] used Johnson's load-sharing concept [31] together with dry circular-contact model by Greenwood and Tripp [32] to investigate the effect of surface roughness without directly solving the Reynolds equation.

In this study, a numerical model is presented to investigate the effect of surface roughness in point-contact (elliptical contact) elastohydrodynamic lubrication. Similar to the authors' approach for line-contact EHL [33–35], the modified Reynolds equation by Patir and Cheng [16], the bulk elastic deformation of the surfaces, and the elasto-plastic asperity contact equations by Zhao et al. [36] are simultaneously solved to obtain the pressure and the film profile. The central and the minimum film thickness and the asperity load ratio are then obtained from the results. At the next step, by performing the regression analyses based on the results of a wide range of simulations, curve-fit expressions are presented to readily predict the film thickness and the asperity load ratio in mixed lubrication elliptical contact. Finally, the results from these formulas are compared with the results presented in other publications to show the efficiency of the present approach.

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Nomenclature

a	half Hertzian length (m)	$\bar{w}_1$	w_1/R_x
b	half Hertzian width (m)	w_1^*	w_1/σ
B	contact length in line contact	w_2	critical interference at the point of fully plastic flow, $54w_1$, (m)
D	R_y/R_x	$\bar{w}_2$	w_2/R_x
E'	effective modulus of elasticity, $1/E' = 0.5[(1-\nu_1^2)/E_1 + (1-\nu_2^2)/E_2]$ (Pa)	w_2^*	w_2/σ
F	normal load (N)	W	dimensionless load number ($F/E'R_x^2$)
G	dimensionless material number ($E'\alpha$)	x	coordinate in moving direction (m)
h	film thickness (m)	y	coordinate perpendicular to moving direction (m)
h^*	h/σ	X	dimensionless coordinate in moving direction (x/b)
h_c	central film thickness (m)	Y	dimensionless coordinate perpendicular to moving direction, (y/a)
$h_{\min}$	minimum film thickness (m)	y_s	distance between the mean line of the surface and the mean line of its summits (m)
h_T	average gap between two surfaces (m)	$\bar{y}_s$	y_s/R_x
H	dimensionless film thickness (h/R_x)	y_s^*	y_s/σ
H_c	dimensionless central film thickness (h_c/R_x)	z	height of asperities measured from the mean line of the summits (m)
$H_{\min}$	dimensionless minimum film thickness, ($h_{\min}/R_x$)	z^*	z/σ
H_T	dimensionless average gap between two surfaces (h_T/R_x)	Z	viscosity–pressure index
L_a	asperity load ratio (%)	α	pressure–viscosity coefficient (m^2/N)
n	asperity density (m^{-2})	β	asperity radius (m)
$\bar{n}$	dimensionless asperity density (nR^2)	$\bar{\beta}$	dimensionless asperity radius (β/R_x)
p	total pressure (Pa)	γ	b/R_x
p_a	asperity pressure (Pa)	κ	ellipticity parameter (a/b)
p_h	hydrodynamic pressure (Pa)	Λ	film parameter ($h_{\min}/\sigma$)
P	dimensionless total pressure	μ	lubricant viscosity, (Pa s)
P_a	dimensionless asperity pressure	μ_0	lubricant viscosity at zero pressure (Pa s)
P_h	dimensionless hydrodynamic pressure	$\bar{\mu}$	dimensionless viscosity (μ/μ_0)
R_x	equivalent contact radius in x direction, $[1/R_{1x} \pm 1/R_{2x}]^{-1}$ (m)	ρ	lubricant density (kg/m^3)
R_y	equivalent contact radius in y direction, $[1/R_{1y} \pm 1/R_{2y}]^{-1}$ (m)	ρ_0	lubricant density at zero pressure (kg/m^3)
u	rolling speed, $(u_1 + u_2)/2$ (m/s)	$\bar{\rho}$	dimensionless density (ρ/ρ_0)
U	dimensionless speed number ($\mu_0 u/E'R_x$)	σ	standard deviation of the surface heights (m)
v	Vickers hardness (Pa)	$\bar{\sigma}$	dimensionless surface roughness (σ/R_x)
V	dimensionless hardness number (v/E')	σ_s	standard deviation of the surface summits (m)
w_1	critical interference at the point of initial yield, $(0.6\pi v/E)^2\beta$ (m)	$\bar{\sigma}_s$	σ_s/R_x
		ϕ_x	pressure flow factor in x direction
		ϕ_y	pressure flow factor in y direction

2. Model

In order to predict the pressure distribution and the film profile in point-contact elastohydrodynamic lubrication (EHL), the Reynolds equation, the bulk deformation of contact surfaces, and the load balance equation should be solved simultaneously. When the surfaces are rough, the Reynolds equation should be modified to include the roughness effects. Moreover, since the load is shared between the fluid and the asperities in the mixed EHL, the roughness changes the load balance. Therefore, the pressure at any point in the contact area is the sum of hydrodynamic and asperity pressures:

$$p = p_h + p_a \quad (1)$$

where p , p_h and p_a are the total, hydrodynamic and asperity pressures, respectively.

Fluid equations: For point-contact EHL, the 2D Reynolds equation modified by Patir and Cheng [16] is written as

$$\frac{\partial}{\partial x} \left(\phi_x \frac{\rho h^3}{12\mu} \frac{\partial p_h}{\partial x} \right) + \frac{\partial}{\partial y} \left(\phi_y \frac{\rho h^3}{12\mu} \frac{\partial p_h}{\partial y} \right) = u \frac{\partial (\rho h_T)}{\partial x} \quad (2)$$

where x and y are the coordinates along and perpendicular to the moving direction, respectively. In Eq. (2), h is the film thickness, μ is the fluid viscosity, ρ is the fluid density, and u is the rolling speed. ϕ_x and ϕ_y are the pressure flow factor in x and y directions and h_T is the average gap between two surfaces [16].

For an isotropic surface with Gaussian distribution of the surface heights, the pressure flow factors are [16]

$$\phi_x = \phi_y = 1 - 0.9e^{-0.56(h/\sigma)} \quad (3)$$

where σ is the surface roughness which is the standard deviation of the surface heights (also shown as R_q in some literature). For Gaussian distribution of the surface heights, h_T can be written as [37]

$$h_T = 0.5h \left[1 + \operatorname{erf} \left(\frac{h}{\sqrt{2}\sigma} \right) \right] + \frac{\sigma}{\sqrt{2\pi}} \exp \left(-\frac{h^2}{2\sigma^2} \right) \quad (4)$$

In Eq. (2), both the viscosity and density are functions of the hydrodynamic pressure. For pressure-density dependence, the equation provided by Dowson and Higginson [38] is used

$$\frac{\rho}{\rho_0} = 1 + \frac{0.6p_h}{1 + 1.7p_h} \quad (5)$$

where ρ_0 is the density at atmospheric pressure. For the viscosity, widely-used Roelands equation [39] which is valid for using at moderate pressures is utilized

$$\frac{\mu}{\mu_0} = \exp \left[(\ln \mu_0 + 9.67) \left\{ -1 + \left(1 + 5.1 \times 10^{-9} p_h \right)^Z \right\} \right] \quad (6)$$

where μ_0 is the viscosity at zero pressure, and Z is viscosity-pressure index which is a function of pressure-viscosity coefficient (α) and μ_0 [40].

The governing equations are non-dimensionalized using the following parameters

$$D = \frac{R_y}{R_x}, \quad X = \frac{x}{b}, \quad Y = \frac{y}{a}, \quad H = \frac{h}{R_x}, \quad \bar{\mu} = \frac{\mu}{\mu_0}, \quad \bar{\rho} = \frac{\rho}{\rho_0}, \quad P = \frac{p}{p_H} \\ U = \frac{\mu_0 U}{E' R_x}, \quad W = \frac{F}{E' R_x^2}, \quad G = \alpha E', \quad \bar{\sigma} = \frac{\sigma}{R_x}, \quad V = \frac{v}{E'} \quad (7)$$

where p is the contact pressure, F is the normal load, E' is the effective modulus of elasticity, and R_x and R_y are the radii of curvature in x and y directions, respectively (See Fig. 1). The parameter v is the Vickers hardness of the softer surface, and parameters U , W , G , $\bar{\sigma}$, and V denote the dimensionless speed, load, material, surface roughness, and surface hardness, respectively.

In Eq. (7), parameters p_H , a , and b are the maximum Hertzian pressure, Hertzian half contact length (along the y direction), and the Hertzian half contact width (along the x direction), respectively. These parameters are defined as

$$b = \sqrt[3]{\frac{6R_x D F \tilde{S}}{\pi E' \kappa (D+1)}} = R_x \sqrt[3]{\frac{6DW\tilde{S}}{\pi \kappa (D+1)}}, \quad a = \kappa b, \quad a \geq b \\ p_H = \frac{3F}{2\pi ab} = \frac{3WE'R_x^2}{2\pi \kappa b^2} = \frac{3WE'}{2\pi \kappa \gamma^2}, \quad \left(\gamma = \frac{b}{R_x} = \sqrt[3]{\frac{6DW\tilde{S}}{\pi \kappa (D+1)}} \right) \quad (8)$$

where κ is the ellipticity parameter defined as the ratio of the Hertzian contact length to the Hertzian contact width (See Fig. 1). This parameter is only a function of D and can be written as:

$$\kappa = \sqrt{(D+1) \frac{\tilde{F}}{\tilde{S}} - D} \quad (9)$$

where $\tilde{F}$ and $\tilde{S}$ are elliptical integral of the first and the second kind

$$\tilde{F}(\kappa) = \int_0^{\pi/2} \frac{d\tau}{\sqrt{1 - (1 - (1/\kappa^2)) \sin^2 \tau}}, \\ \tilde{S}(\kappa) = \int_0^{\pi/2} \sqrt{1 - (1 - (1/\kappa^2)) \sin^2 \tau} d\tau \quad (10)$$

It should be noted that since the elliptical integrals are functions of the ellipticity, an iterative procedure is needed to calculate κ based on D .

The Reynolds equation can be written in dimensionless form

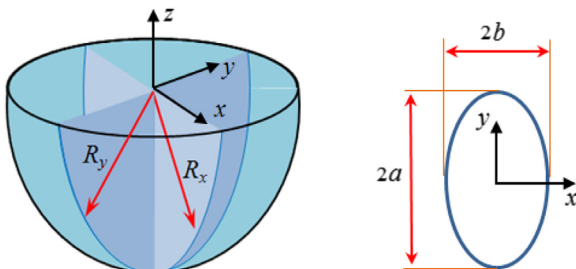


Fig. 1. Coordinates and geometry of elliptical contact.

as

$$\frac{\partial}{\partial X} \left(\phi_X \frac{\bar{\rho} H^3}{\bar{\mu}} \frac{\partial P_h}{\partial X} \right) + \frac{1}{\kappa^2} \frac{\partial}{\partial Y} \left(\phi_Y \frac{\bar{\rho} H^3}{\bar{\mu}} \frac{\partial P_h}{\partial Y} \right) = \frac{8\pi \kappa \gamma^3 U}{W} \frac{\partial (\bar{\rho} H_T)}{\partial X} \quad (11)$$

where H_T is the dimensionless average gap, and ϕ_X and ϕ_Y are the isotropic flow factors written as functions of dimensionless parameters as

$$\phi_X = \phi_Y = 1 - 0.9e^{-0.56(H/\bar{\sigma})} \quad (12)$$

Load balance: The total load is obtained by integrating the hydrodynamic and the asperity pressures within the contact region

$$F = \iint p_h(x, y) dx dy + \iint p_a(x, y) dx dy \quad (13)$$

Eq. (13) is written in dimensionless form as

$$\frac{2\pi}{3} = \iint P_h(X, Y) dX dY + \iint P_a(X, Y) dX dY \quad (14)$$

Bulk deformation and film profile: The lubricant film has the following form considering Hertzian geometry and elastic deformation of the surfaces according to [41]:

$$h(x, y) = h_{00} + \frac{x^2}{2R_x} + \frac{y^2}{2R_y} + \frac{2}{\pi E'} \iint \frac{p(s, q) ds dq}{\sqrt{(x-s)^2 + (y-q)^2}} \quad (15)$$

where h_{00} is a constant. Physically, the load balance adjusts the separation between the surfaces which appears in this constant. It should be noted that the surface deformation is caused by the total pressure. Eq. (15) can be nondimensionalized as

$$H(X, Y) = H_{00} + \gamma^2 \frac{X^2}{2} + \frac{\gamma^2 \kappa^2 Y^2}{2} + \frac{\gamma^2 \kappa (D+1)}{2\pi \tilde{S}} \iint \frac{P(S, Q) dS dQ}{\sqrt{(X-S)^2 + \kappa^2 (Y-Q)^2}} \quad (16)$$

where H_{00} is a constant to be determined.

Surface roughness: In this study, for the deformation of surface asperities, the elasto-plastic model known as ZMC developed by Zhao et al. [36] is utilized which considers the elastic, elasto-plastic and fully plastic deformation of the asperities. Recently, Beheshti and Khonsari [42] showed that the ZMC model gives close results compared to other comprehensive asperity micro-contact models. This model has also been utilized in recent studies on the mixed EHL [33–35,43]. In this model, the asperity pressure P_a is defined as the sum of elastic, elasto-plastic and plastic pressures:

$$p_a = \frac{2}{3} E' n \beta^{0.5} \sigma^{1.5} \left(\frac{\sigma}{\sigma_s} \right) \frac{1}{\sqrt{2\pi}} \int_{h^* - y_s^* + w_1}^{h^* - y_s^* + w_1^*} w^{*1.5} e^{-0.5((\sigma/\sigma_s)z^*)^2} dz^* \\ + 2\pi v n \beta \sigma \frac{1}{\sqrt{2\pi}} \left(\frac{\sigma}{\sigma_s} \right) \int_{h^* - y_s^* + w_2}^{\infty} w^* e^{-0.5((\sigma/\sigma_s)z^*)^2} dz^* \\ + \pi v n \beta \sigma \frac{1}{\sqrt{2\pi}} \left(\frac{\sigma}{\sigma_s} \right) \int_{h^* - y_s^* + w_1}^{h^* - y_s^* + w_2^*} w^* e^{-0.5((\sigma/\sigma_s)z^*)^2} dz^*$$

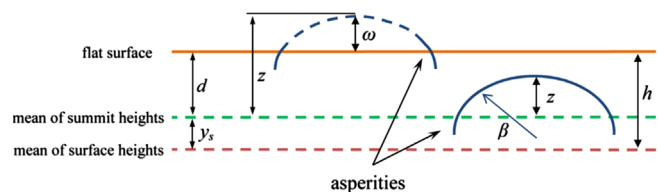


Fig. 2. Contact between a rough and a smooth surface.

$$\times \left[1 - 0.6 \frac{\ln w_2^* - \ln w_1^*}{\ln w_2^* - \ln w_1^*} \right] \left[1 - 2 \left(\frac{w_2^* - w_1^*}{w_2^* - w_1^*} \right)^3 + 3 \left(\frac{w_2^* - w_1^*}{w_2^* - w_1^*} \right)^2 \right] dz^* \quad (17)$$

where $w^* = z^* - h^* + y_s^*$. In Eq. (17), the starred variables are normalized by σ . Note that σ is the standard deviation of the surface heights, while σ_s is the standard deviation of the surface summits. Parameter h is the separation (film thickness in EHL contact), β is the asperity radius, and n is the asperity density. Fig. 2 shows a schematic view of the contact between a rough and a smooth surface.

Eq. (17) can be written in dimensionless form as

$$P_a = \frac{2\pi\gamma^2\kappa}{3WE} P_a = \frac{4\pi\gamma^2\kappa}{9W} \bar{n}\bar{\beta}\bar{\sigma}^{0.5} \bar{\sigma}^{-1.5} \left(\frac{\bar{\sigma}}{\bar{\sigma}_s} \right) \int_{I_1}^{I_2} (z^* - I_1)^{1.5} e^{-0.5((\bar{\sigma}/\bar{\sigma}_s)z^*)^2} dz^* \\ + \frac{4\pi^2\gamma^2\kappa}{3W} V\bar{n}\bar{\beta}\bar{\sigma} \left(\frac{\bar{\sigma}}{\bar{\sigma}_s} \right) \int_{I_3}^{\infty} (z^* - I_1) e^{-0.5((\bar{\sigma}/\bar{\sigma}_s)z^*)^2} dz^* \\ + \frac{2\pi^2\gamma^2\kappa}{3W} V\bar{n}\bar{\beta}\bar{\sigma} \left(\frac{\bar{\sigma}}{\bar{\sigma}_s} \right) \int_{I_2}^{I_3} (z^* - I_1) e^{-0.5((\bar{\sigma}/\bar{\sigma}_s)z^*)^2} dz^* \\ \times \left[1 - 0.6 \frac{\ln \bar{w}_2 - \ln(\bar{\sigma}(z^* - I_1))}{\ln \bar{w}_2 - \ln \bar{w}_1} \right] \left[1 - 2 \left(\frac{\bar{\sigma}(z^* - I_1) - \bar{w}_1}{\bar{w}_2 - \bar{w}_1} \right)^3 \right. \\ \left. + 3 \left(\frac{\bar{\sigma}(z^* - I_1) - \bar{w}_1}{\bar{w}_2 - \bar{w}_1} \right)^2 \right] dz^* \quad (18)$$

where $I_1 = \frac{H - \bar{y}_s}{\bar{\sigma}}$, $I_2 = \frac{H - \bar{y}_s + \bar{w}_1}{\bar{\sigma}}$, $I_3 = \frac{H - \bar{y}_s + \bar{w}_2}{\bar{\sigma}}$.

3. Numerical simulation procedure

The governing Eqs. (11), (14), (16) and (18) are discretized using the finite difference method and solved simultaneously to obtain the pressure and the film profiles. The finite difference formulation for the Reynolds equation is provided in Appendix A. The dimensionless input parameters are the load W , speed U , material G , surface roughness $\bar{\sigma}$, surface hardness V , and ellipticity parameter κ . It should be noted that for the sake of minimizing the number of input parameters, the asperity radius β and the asperity density n are omitted from the input by the assumption of $n\beta\sigma=0.05$ [31,33,44] and $\sigma/\beta=0.01$ [33]. Also, when both surfaces are rough, the combined roughness of $\sigma = \sqrt{\sigma_1^2 + \sigma_2^2}$ should be used.

For the line-contact EHL problem, the widely-used solution method is the Newton–Raphson algorithm. The convergence rate of this method is fast (quadratic), but the procedure requires decomposition of the Jacobian matrix to solve the equations. For 1-D problem of line-contact, applying Newton–Raphson algorithm was found to be the best choice, while the use of this approach for 2-D problem of point contact is not efficient. In other words, for N nodes along the line contact domain, the number of elements in the Jacobian matrix is equal to N^2 while for point contact problem with $N \times N$ number of nodes, the Jacobian matrix has N^4 elements, which is difficult to handle.

The widely-used method for solving the point-contact EHL problem is relaxation method which is also utilized in this study. In fact, Successive Over-relaxation Method (S.O.R) is used to solve the discretized Reynolds equation in an inner loop, while the hydrodynamic and asperity pressures are updated by an under-relaxation factor (starts from 0.01 and decreases gradually throughout the process) in an outer loop. The other unknown parameter H_{00} is also under-relaxed by the load balance in the outer loop as well. Appendix B shows a flowchart of the numerical procedure.

To ensure that the fully flooded condition prevails, the minimum amount of X is set to -4 in all cases simulated. Appendix C shows the effect of inlet variation on the results for both smooth

and rough surfaces. Also, while the hydrodynamic pressure becomes nil at a point close to $X=1$, the asperity pressure may have a value far beyond that point and thus the maximum amount of X is set to 2. Choosing the limits of Y as ± 2 , the computational domain is a rectangle with $-4 \leq X \leq 2$ and $-2 \leq Y \leq 2$. The hydrodynamic pressure is zero along all boundaries. Also, the hydrodynamic pressure is set to zero whenever it becomes negative (commonly known as Reynolds condition [1]). It should be mentioned that since the problem is symmetric with respect to the X axis ($Y=0$ plane), only half of the problem is modeled and solved. Therefore, the actual computational domain is $-4 \leq X \leq 2$ and $0 \leq Y \leq 2$ with symmetric boundary conditions on $Y=0$ plane.

The convergence of the results is ensured by refining the mesh until the change in the results becomes negligible. In general, within the range of the operating conditions studied, 30 nodes in one unit along the X direction and 20 nodes in one unit along the Y direction (which results in a 180×40 mesh for $-4 \leq X \leq 2$ and $0 \leq Y \leq 2$) is proven to be sufficient. Appendix D presents the results of a mesh refinement study for both the smooth and rough surfaces. As shown, the error uniformly decreases as the computational mesh is refined. However, further refinement beyond 180×40 nodes yields less than 1% error at the expense of much greater computational demand.

The run time of the simulations is dependent on the input parameters. In general, solution for the higher loads takes longer because more iteration is needed to converge. Also, the solution for rough surfaces is generally slower than the smooth solution, because the asperity pressure must be calculated in each loop. The typical run time on an Intel i7 3.47 GHz machine for the chosen domain and grid setup is generally between 20 and 100 minutes depending on the input.

The final results consist of the hydrodynamic, asperity, and total pressure distributions as well as the film profile. Based on these results, the central and the minimum film thickness are easily determined. Also predicted is the asperity load ratio which is the ratio of the load carried by the asperities to the total normal load.

4. Results and discussions

Fig. 3 shows a typical example of 3D distribution of dimensionless pressure in a rough EHL problem. Since a part of the load is carried by the surface asperities, the total pressure is composed of hydrodynamic and asperity parts. For the case depicted in Fig. 3, the simulation yields the asperity load ratio of 54%, which means that the volume under the asperity pressure is 54% of that of the total pressure.

Fig. 4 depicts an example of contour plots for the dimensionless film thickness distribution in smooth and rough surfaces. As shown, for both cases, the minimum film thickness occurs at the sides (Point B). However, when dealing with rough surfaces, this point is closer to the center in the Y direction. For the smooth case, the simulations predict a central film thickness of $H_c = 1.80 \times 10^{-5}$ (Point A) and a minimum film thickness of $H_{\min} = 1.05 \times 10^{-5}$ (point B). For the rough surface, the central film thickness is $H_c = 2.31 \times 10^{-5}$ (Point A) and the minimum film thickness is $H_{\min} = 1.82 \times 10^{-5}$ (point B). Note that the results are symmetric with respect to $Y=0$ plane.

To have a better understanding of the results, the pressure and the film distributions are plotted along the X direction in $Y=0$ plane and along the Y direction in $X=0$ plane in Fig. 5. For all cases, the input are $\kappa=1$, $W=1 \times 10^{-6}$, $U=1 \times 10^{-11}$ and $G=4972$. Also, the dimensionless hardness is set to $V=0.01$ (Vickers hardness of 2.35 GPa equal to 20 Rockwell C, which is a reasonable value for regular steel). The results are presented for three

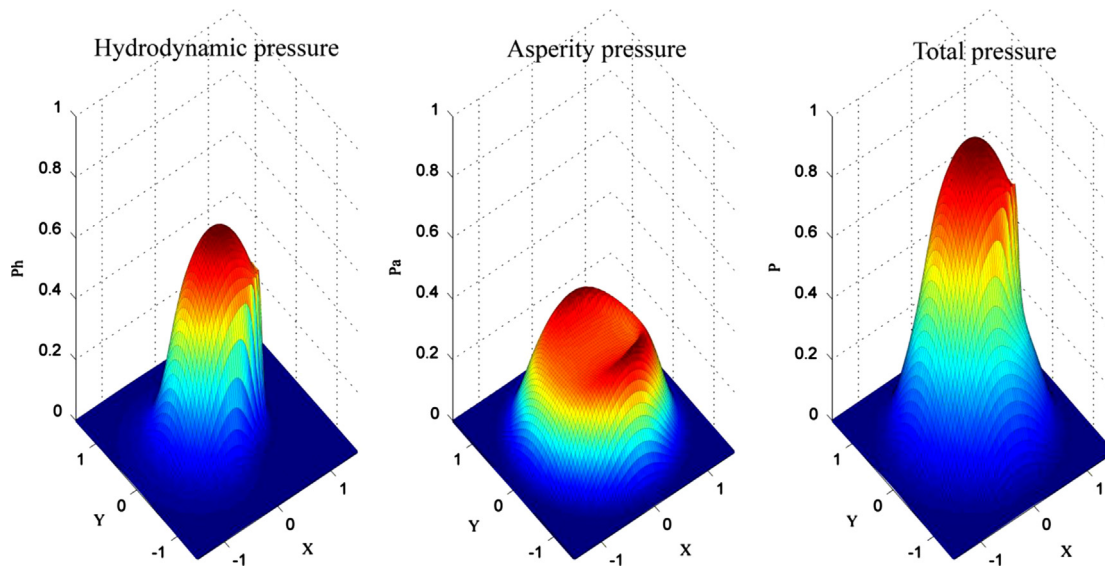


Fig. 3. Hydrodynamic, asperity and total pressure distribution of rough surfaces ($\kappa=1$, $W=1 \times 10^{-6}$, $U=1 \times 10^{-11}$, $G=4972$, $\bar{\sigma}=3 \times 10^{-5}$, and $V=0.01$).

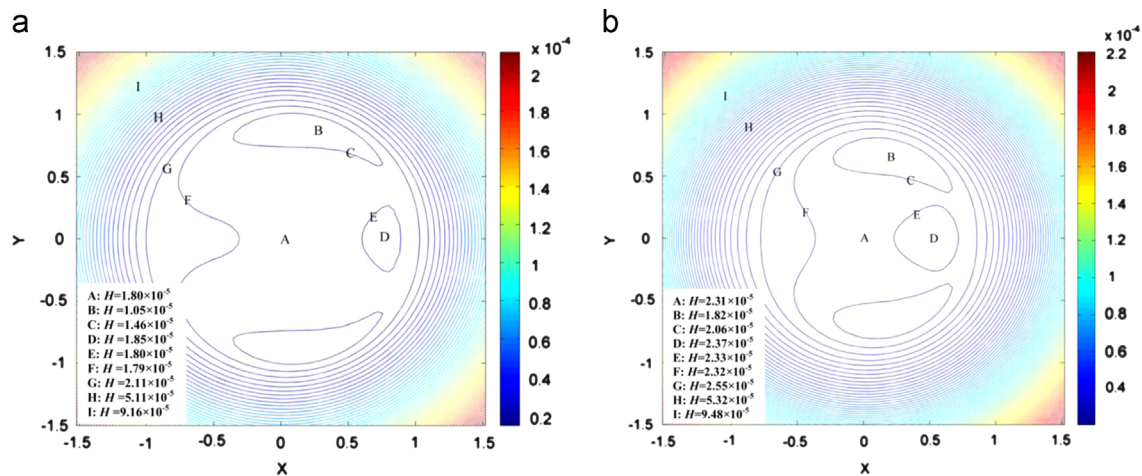


Fig. 4. Film thickness distributions ($\kappa=1$, $W=1 \times 10^{-6}$, $U=1 \times 10^{-11}$, $G=4972$) (a) smooth surfaces and (b) rough surfaces with $\bar{\sigma}=3 \times 10^{-5}$.

roughness levels: a smooth surface (typically $\bar{\sigma}=1 \times 10^{-6}$ satisfies this assumption), slightly rough ($\bar{\sigma}=1 \times 10^{-5}$) and fairly rough ($\bar{\sigma}=3 \times 10^{-5}$). It is obvious that there is no asperity pressure for the smooth case ($L_a=0$). The predicted asperity load ratio is $L_a=8\%$ for $\bar{\sigma}=1 \times 10^{-5}$ and $L_a=54\%$ for $\bar{\sigma}=3 \times 10^{-5}$.

As shown in Fig. 5(a), when dealing with rough surfaces, total pressure extends beyond $X=1$ due to the existence of asperity pressure there, while the hydrodynamic pressure becomes nil around $X=1$. This implies that the separation gap beyond the width of the Hertzian contact is small enough to cause some of the asperities to experience intimate contact. This phenomenon is, of course, more pronounced for the highest roughness values simulated.

As Fig. 5(b) shows, the pressure profiles are symmetric with respect to $Y=0$ plane. Also, the asperity pressure profiles have a peak at each side which is due to local decrease in the film thickness. Also, for the case of rough surfaces, the asperity pressure exists at the sides beyond the points where the hydrodynamic pressure becomes negligible. This is again more visible for the highest roughness value and affects the total pressure distribution.

Fig. 6 show the film thickness distributions along the X axis in $Y=0$ plane and along the Y axis in $X=0$ plane. The results for smooth surfaces are compared with those for rough surfaces ($\bar{\sigma}=1 \times 10^{-5}$ and $\bar{\sigma}=3 \times 10^{-5}$). As shown, increasing the surface roughness increases the film thickness. This is due to the contribution of the asperities in carrying the contact load (which decreases the part carried by the fluid) as well as their influence on the flow behavior (which is accounted for by the flow factors in the modified Reynolds equation). However, increasing the roughness decreases the film parameter Λ which is the ratio of the minimum film thickness to the combined surface roughness ($\Lambda=h_{\min}/\sqrt{\sigma_1^2+\sigma_2^2}$). This parameter is $\Lambda=1.3$ for $\bar{\sigma}=1 \times 10^{-5}$ and $\Lambda=0.6$ for $\bar{\sigma}=3 \times 10^{-5}$.

It is worthwhile to mention that while the film thickness value at the point $X=0$ in Fig. 6(a) and $Y=0$ in Fig. 6(b) represent the central film thickness, the minimum film thickness does not appear in either of the views. This is because the minimum value of the film thickness occurs at the sides which is not located in $X=0$ or $Y=0$ plane (See Fig. 4). Also note that as the roughness increases, the film profiles shrink toward the center in both X and Y directions.

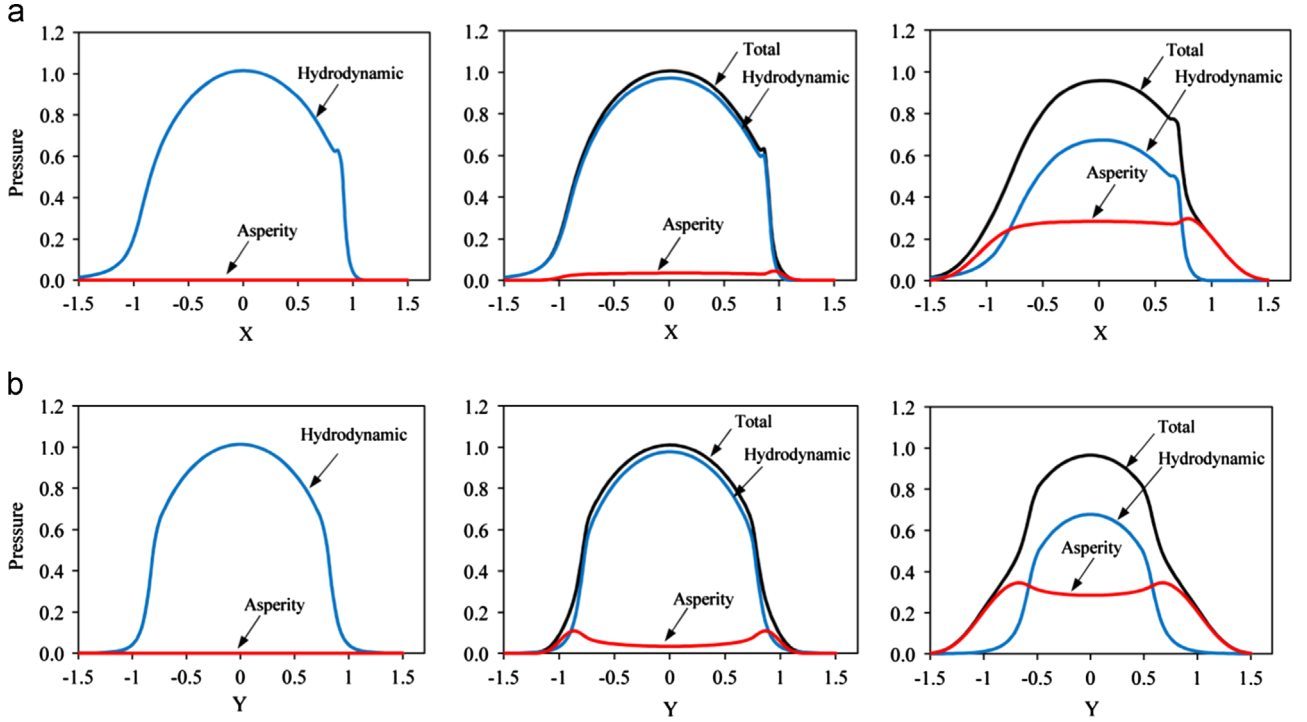


Fig. 5. Pressure distributions: (a) along X in $Y=0$ plane and (b) along Y in $X=0$ plane ($\kappa=1$, $W=1 \times 10^{-6}$, $U=1 \times 10^{-11}$, $G=4972$, and $V=0.01$) From left to right: smooth surface, rough surface with $\bar{\sigma}=1 \times 10^{-5}$, rough surface with $\bar{\sigma}=3 \times 10^{-5}$.

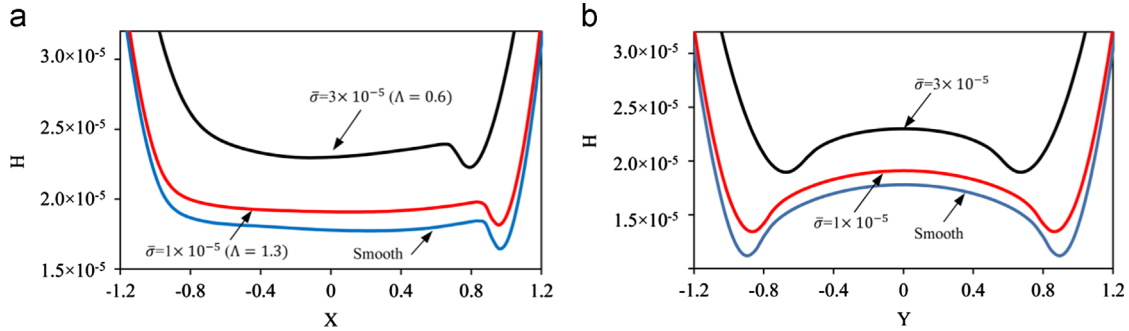


Fig. 6. Film profiles ($\kappa=1$, $W=1 \times 10^{-6}$, $U=1 \times 10^{-11}$, $G=4972$, and $V=0.01$) (a) along X in $Y=0$ plane and (b) along Y in $X=0$ plane.

5. Predictive formulas

The main goal of this study is to quantify the effect of surface roughness in point-contact EHL. This is done by curve-fitting the results of a set of simulations using a wide range of input. In this section, we first present the smooth results for comparison and benchmarking purpose. In the next step, rough surface results are presented and predictive formulas are derived for the central and minimum film thickness as well as the asperity load ratio in rough point-contact EHL.

5.1. Film thickness formulas for smooth point-contact EHL

The input parameters for the smooth point-contact EHL simulations are ellipticity κ , dimensionless load W , dimensionless speed U , and dimensionless material G . Table 1 shows the range of input parameters selected for the simulations.

A reasonable form of curve-fit equation for the central and the minimum film thickness can be assumed as $H=c_1 W^{c_2} U^{c_3} G^{c_4} (1+c_5 e^{c_6 \kappa})$ which was used by Hamrock and Dowson [3]. Since changing the ellipticity affects the contact characteristics, we implement a new form in which the exponents of W , U , and G are also assumed to be functions of ellipticity. This form also

Table 1

Range of input parameters selected for simulation (smooth surfaces).

Parameter	W	U	G	κ
Min	1.5×10^{-7}	1×10^{-12}	2500	1
Max	2.5×10^{-4}	1×10^{-10}	7500	8

accommodates a wider range of load. Therefore, a general form of $H=c_1 W^{c_2 \kappa^{m_1}} U^{c_3 \kappa^{m_2}} G^{c_4 \kappa^{m_3}} (1+c_5 e^{c_6 \kappa})$ is chosen for performing the regression analysis. After analyzing the results of about a hundred simulations within the range of input shown in Table 1, the curve-fit equations for the central and minimum film thickness (smooth surface) are obtained as

$$(H_c)_{\text{smooth}} = \frac{(h_c)_{\text{smooth}}}{R_x} = 3.672W^{-0.045\kappa^{0.18}} U^{0.663\kappa^{0.025}} G^{0.502\kappa^{0.064}} (1-0.573e^{-0.74\kappa}) \quad (19)$$

$$(H_{\min})_{\text{smooth}} = \frac{(h_{\min})_{\text{smooth}}}{R_x} = 1.637W^{-0.09\kappa^{-0.15}} U^{0.711\kappa^{-0.023}} G^{0.65\kappa^{-0.045}} (1-0.974e^{-0.676\kappa}) \quad (20)$$

Table 2
Results of simulations and curve-fit equations (smooth surfaces).

Input				H_c (smooth surface)			H_{min} (smooth surface)		
κ	W	U	G	Simulation	Eq. (19)	Error (%)	Simulation	Eq. (20)	Error (%)
1	1×10^{-6}	1×10^{-11}	4972	1.80×10^{-5}	1.81×10^{-5}	0.56	1.05×10^{-5}	1.09×10^{-5}	3.81
2				2.11×10^{-5}	2.14×10^{-5}	1.42	1.62×10^{-5}	1.61×10^{-5}	0.62
3				2.25×10^{-5}	2.30×10^{-5}	2.22	1.87×10^{-5}	1.88×10^{-5}	0.53
4				2.35×10^{-5}	2.38×10^{-5}	1.28	2.01×10^{-5}	2.02×10^{-5}	0.50
6				2.42×10^{-5}	2.45×10^{-5}	1.24	2.03×10^{-5}	2.14×10^{-5}	5.42
8				2.48×10^{-5}	2.48×10^{-5}	0.00	2.07×10^{-5}	2.19×10^{-5}	5.80
1	1.5×10^{-7}	1×10^{-11}	4972	1.90×10^{-5}	1.98×10^{-5}	4.21	1.33×10^{-5}	1.30×10^{-5}	2.26
	3×10^{-7}			1.88×10^{-5}	1.92×10^{-5}	2.13	1.23×10^{-5}	1.22×10^{-5}	0.81
	2×10^{-6}			1.74×10^{-5}	1.76×10^{-5}	1.15	9.67×10^{-6}	1.03×10^{-5}	6.51
	5×10^{-6}			1.71×10^{-5}	1.69×10^{-5}	1.17	9.42×10^{-6}	9.46×10^{-6}	0.42
	2×10^{-5}			1.74×10^{-5}	1.59×10^{-5}	8.62	8.66×10^{-6}	8.35×10^{-6}	3.58
1	1×10^{-6}	1×10^{-12}	4972	3.69×10^{-6}	3.94×10^{-6}	6.78	1.94×10^{-6}	2.13×10^{-6}	9.79
		5×10^{-12}		1.15×10^{-5}	1.15×10^{-5}	0.00	6.30×10^{-6}	6.68×10^{-6}	6.03
		2×10^{-11}		2.86×10^{-5}	2.87×10^{-5}	0.35	1.79×10^{-5}	1.79×10^{-5}	0.00
		5×10^{-11}		5.21×10^{-5}	5.28×10^{-5}	1.34	3.60×10^{-5}	3.43×10^{-5}	4.72
		1×10^{-10}		8.08×10^{-5}	8.35×10^{-5}	3.34	5.97×10^{-5}	5.62×10^{-5}	5.86
1	1×10^{-6}	1×10^{-11}	2500	1.27×10^{-5}	1.29×10^{-5}	1.57	6.77×10^{-6}	6.99×10^{-6}	3.25
			7500	2.27×10^{-5}	2.23×10^{-5}	1.76	1.44×10^{-5}	1.43×10^{-5}	0.69
	2×10^{-6}	2×10^{-11}	2500	1.87×10^{-5}	1.97×10^{-5}	5.35	1.02×10^{-5}	1.08×10^{-5}	5.88
	6.67×10^{-7}	6.67×10^{-12}	7500	1.75×10^{-5}	1.74×10^{-5}	0.57	1.10×10^{-5}	1.11×10^{-5}	0.91
6	1.5×10^{-6}	1×10^{-11}	4972	2.39×10^{-5}	2.39×10^{-5}	0.00	2.04×10^{-5}	2.08×10^{-5}	1.96
	5×10^{-6}			2.19×10^{-5}	2.21×10^{-5}	0.91	1.94×10^{-5}	1.92×10^{-5}	1.03
	1×10^{-5}			2.04×10^{-5}	2.12×10^{-5}	3.92	1.89×10^{-5}	1.83×10^{-5}	3.17
	5×10^{-5}			1.90×10^{-5}	1.92×10^{-5}	1.05	1.79×10^{-5}	1.64×10^{-5}	8.38
	1×10^{-4}			1.89×10^{-5}	1.84×10^{-5}	2.65	1.71×10^{-5}	1.56×10^{-5}	8.77
6	5×10^{-6}	1×10^{-12}	4972	4.35×10^{-6}	4.49×10^{-6}	3.22	3.91×10^{-6}	3.98×10^{-6}	1.79
		5×10^{-12}		1.36×10^{-5}	1.37×10^{-5}	0.74	1.23×10^{-5}	1.19×10^{-5}	3.25
		2×10^{-11}		3.48×10^{-5}	3.58×10^{-5}	2.87	3.13×10^{-5}	3.08×10^{-5}	1.60
		5×10^{-11}		6.52×10^{-5}	6.76×10^{-5}	3.68	5.92×10^{-5}	5.75×10^{-5}	2.87
		1×10^{-10}		1.03×10^{-4}	1.09×10^{-4}	5.54	9.44×10^{-5}	9.23×10^{-5}	2.22
6	5×10^{-6}	1×10^{-11}	2500	1.52×10^{-5}	1.50×10^{-5}	1.32	1.25×10^{-5}	1.27×10^{-5}	1.60
			7500	2.71×10^{-5}	2.79×10^{-5}	2.95	2.57×10^{-5}	2.45×10^{-5}	4.67
	1×10^{-5}	2×10^{-11}	2500	2.35×10^{-5}	2.33×10^{-5}	0.85	1.94×10^{-5}	1.94×10^{-5}	0.00
	3.33×10^{-6}	6.67×10^{-12}	7500	2.11×10^{-5}	2.16×10^{-5}	2.37	1.96×10^{-5}	1.91×10^{-5}	2.55

Table 2 shows a part of simulation results for the central and the minimum film thickness within the range of input shown in Table 1. In this table, the central and minimum film thickness values obtained from the curve-fit equations (Eqs. (19) and (20)) are compared with those from simulations, and the error between them is shown as well. The average value of the error between simulation and curve-fit results is 2.27% with the maximum of 8.62% for the central film thickness, while it is 3.27% with the maximum of 9.79% for the minimum film thickness.

It should be noted that for a given dimensionless load number W , increasing the ellipticity parameter κ decreases the Hertzian pressure (See Eq. (8)) due to a larger contact area. For example, the dimensionless load $W=1 \times 10^{-5}$ yields a maximum Hertzian pressure of 1.8 GPa (for steel) at $\kappa=1$, but 0.77 GPa at $\kappa=8$. In this study, the dimensionless load value is chosen in a way that keeps the maximum Hertzian pressure between about 0.4 GPa and 2.3 GPa (for steel). So, while the upper limit is obtained as $W=2.5 \times 10^{-4}$ for $\kappa=8$, it is $W=2 \times 10^{-5}$ for $\kappa=1$. Therefore, special attention should be paid while choosing the upper limit of dimensionless load for the use in Eqs. (19) and (20).

5.2. Film thickness and asperity load formulas for rough point-contact EHL

In order to obtain curve-fit expressions for the film thickness and asperity load ratio in point-contact EHL of rough surfaces, an extensive set of simulations are performed based on the approach presented in this study. The range of input parameters selected for the simulations are shown in Table 3. The range of dimensionless load, speed, material and ellipticity parameters are same as those

Table 3
Range of input parameters selected for simulation (rough surfaces).

Parameter	W	U	G	κ	$\bar{\sigma}$	V
Min	1.5×10^{-7}	1×10^{-12}	2500	1	0	0.005
Max	2.5×10^{-4}	1×10^{-10}	7500	8	5×10^{-5}	0.03

selected for smooth simulations (Table 1). The selected range for dimensionless roughness $\bar{\sigma}$ starts from $\bar{\sigma}=0$ (theoretically smooth surface) up to $\bar{\sigma}=5 \times 10^{-5}$ (equal to combined roughness of 1.27 μm for equivalent radius of one inch). However, the results for roughness values below $\bar{\sigma}=1 \times 10^{-6}$ are proved to be very close to the smooth surface solution results. The selected range for the dimensionless hardness V covers the Vickers hardness of about 1.1 to 6.8 GPa for the steel. The upper range equals to about 60 Rockwell C which corresponds to that of most hardened steels. It should also be noted that the allowable range of dimensionless load W should be checked for each ellipticity value as mentioned in the previous section.

The form of the curve-fit expressions for both central and the minimum film thickness are assumed as $H=(H)_{\text{smooth}} \times (1+c_1 W^{c_2} U^{c_3} G^{c_4} \bar{\sigma}^{c_5} V^{c_6} \kappa^{c_7})$ which is similar to the form used for line-contact formulas in Ref. [33]. In this form, when the surface roughness is zero (or practically very small), the film thickness equations revert back to those for smooth surface. The following expressions are obtained by performing the regression analysis for the results of about a hundred simulations within the range of input shown in Table 3 (a total of about 200 simulations for both smooth and rough cases).

Table 4
Results of simulations and curve-fit equations (rough surfaces).

Input						Central film thickness (H_c)			Film parameter (Λ)					
κ	W	U	G	$\bar{\sigma}$	V	Simulation	Eq. (21)	Error (%)						
(a) Central film thickness														
1	1×10^{-6}	1×10^{-11}	4972	2×10^{-5}	0.01	2.13×10^{-5}	2.11×10^{-5}	0.94	0.80					
2						2.51×10^{-5}	2.50×10^{-5}	0.40	1.08					
3						2.63×10^{-5}	2.70×10^{-5}	2.66	1.22					
4						2.74×10^{-5}	2.80×10^{-5}	2.19	1.28					
6						2.86×10^{-5}	2.90×10^{-5}	1.40	1.31					
8						2.93×10^{-5}	2.95×10^{-5}	0.68	1.31					
1	1.5×10^{-7}	1×10^{-11}	4972	2×10^{-5}	0.01	2.26×10^{-5}	2.39×10^{-5}	5.75	0.97					
						5×10^{-6}		1.91×10^{-5}	1.91×10^{-5}	0.00	0.67			
						2×10^{-5}		1.86×10^{-5}	1.76×10^{-5}	5.38	0.59			
1						1×10^{-6}	5×10^{-12}	4972	2×10^{-5}	0.01	1.47×10^{-5}	1.49×10^{-5}	1.36	0.60
											3×10^{-11}		3.97×10^{-5}	3.99×10^{-5}
	1×10^{-10}		8.16×10^{-5}	8.53×10^{-5}	4.53						3.00			
1	2×10^{-6}	2×10^{-11}	2500	2×10^{-5}	0.01						2.24×10^{-5}	2.28×10^{-5}	1.79	0.80
											6.67×10^{-7}	6.67×10^{-12}	7500	2.08×10^{-5}
						1×10^{-6}	1×10^{-11}	2500	1.66×10^{-5}	1.69×10^{-5}	1.81	0.66		
								7500	2.56×10^{-5}	2.47×10^{-5}	3.52	0.93		
6						5×10^{-6}	1×10^{-11}	4972	2×10^{-5}	0.01	2.51×10^{-5}	2.55×10^{-5}	1.59	1.16
			2.38×10^{-5}	2.41×10^{-5}	1.26						1.10			
			2.01×10^{-5}	2.02×10^{-5}	0.50						0.92			
6	5×10^{-6}	5×10^{-12}	4972	2×10^{-5}	0.01						1.76×10^{-5}	1.75×10^{-5}	0.57	0.82
											3×10^{-11}		4.82×10^{-5}	5.01×10^{-5}
						1×10^{-10}		1.04×10^{-4}	1.11×10^{-4}	6.73	4.76			
6						1×10^{-5}	2×10^{-11}	2500	2×10^{-5}	0.01	2.71×10^{-5}	2.67×10^{-5}	1.48	1.19
											3.33×10^{-6}	6.67×10^{-12}	7500	2.48×10^{-5}
	5×10^{-6}	1×10^{-11}	2500	1.99×10^{-5}	1.95×10^{-5}						2.01	0.89		
			7500	3.01×10^{-5}	3.07×10^{-5}						1.99	1.40		
1	1×10^{-6}	1×10^{-11}	4972	1×10^{-6}	0.01						1.80×10^{-5}	1.82×10^{-5}	1.11	11.10
						5×10^{-6}		1.82×10^{-5}	1.87×10^{-5}	2.75	2.40			
						1×10^{-5}		1.91×10^{-5}	1.94×10^{-5}	1.57	1.33			
						3×10^{-5}		2.31×10^{-5}	2.30×10^{-5}	0.43	0.62			
6						5×10^{-6}	1×10^{-11}	4972	4×10^{-5}	0.01	2.46×10^{-5}	2.51×10^{-5}	2.03	0.53
	5×10^{-6}		2.20×10^{-5}	2.27×10^{-5}	3.18						4.02			
	1×10^{-5}		2.22×10^{-5}	2.35×10^{-5}	5.86						2.11			
	3×10^{-5}		2.79×10^{-5}	2.77×10^{-5}	0.72						0.84			
	4×10^{-5}		2.98×10^{-5}	3.00×10^{-5}	0.67						0.69			
			5×10^{-5}			3.20×10^{-5}	3.26×10^{-5}	1.88	0.60					
1	1×10^{-6}	1×10^{-11}	4972	2×10^{-5}	0.005	2.09×10^{-5}	2.08×10^{-5}	0.48	0.77					
						0.02	2.14×10^{-5}	2.13×10^{-5}	0.47	0.82				
						0.03	2.15×10^{-5}	2.15×10^{-5}	0.00	0.84				
6						5×10^{-6}	1×10^{-11}	4972	2×10^{-5}	0.005	2.46×10^{-5}	2.52×10^{-5}	2.44	1.14
											0.02	2.52×10^{-5}	2.58×10^{-5}	2.38
	0.03	2.54×10^{-5}	2.59×10^{-5}	1.97	1.20									
Input											Minimum film thickness ($H_{\min}$)			Film parameter (Λ)
κ	W	U	G	$\bar{\sigma}$	V						Simulation	Eq. (22)	Error (%)	
(b) Minimum film thickness														
1	1×10^{-6}	1×10^{-11}	4972	2×10^{-5}	0.01	1.58×10^{-5}	1.59×10^{-5}	0.63	0.80					
2						2.15×10^{-5}	2.16×10^{-5}	0.47	1.08					
3						2.37×10^{-5}	2.43×10^{-5}	2.53	1.22					
4						2.43×10^{-5}	2.55×10^{-5}	4.94	1.28					
6						2.52×10^{-5}	2.62×10^{-5}	3.97	1.31					
8						2.58×10^{-5}	2.62×10^{-5}	1.55	1.31					
1	1.5×10^{-7}	1×10^{-11}	4972	2×10^{-5}	0.01	1.93×10^{-5}	1.93×10^{-5}	0.00	0.97					

Table 4 (continued)

Input						Minimum film thickness ($H_{\min}$)			Film parameter (Λ)
κ	W	U	G	$\bar{\sigma}$	V	Simulation	Eq. (22)	Error (%)	
1	5×10^{-6} 2×10^{-5} 1×10^{-6}	5×10^{-12} 3×10^{-11} 1×10^{-10}	4972	2×10^{-5}	0.01	1.30×10^{-5}	1.34×10^{-5}	3.08	0.67
						1.16×10^{-5}	1.17×10^{-5}	0.86	0.59
						1.13×10^{-5}	1.20×10^{-5}	6.19	0.60
						2.86×10^{-5}	2.82×10^{-5}	1.40	1.41
1	2×10^{-6} 6.67×10^{-7} 1×10^{-6}	2×10^{-11} 6.67×10^{-12} 1×10^{-11}	2500	2×10^{-5}	0.01	6.12×10^{-5}	6.00×10^{-5}	1.96	3.00
						1.54×10^{-5}	1.59×10^{-5}	3.25	0.80
						1.62×10^{-5}	1.59×10^{-5}	1.85	0.80
						1.25×10^{-5}	1.31×10^{-5}	4.80	0.66
6	5×10^{-6} 1×10^{-5} 1×10^{-4} 5×10^{-6}	1×10^{-11}	4972	2×10^{-5}	0.01	1.91×10^{-5}	1.86×10^{-5}	2.62	0.93
						2.35×10^{-5}	2.31×10^{-5}	1.70	1.16
						2.31×10^{-5}	2.20×10^{-5}	4.76	1.10
						2.00×10^{-5}	1.84×10^{-5}	8.00	0.92
6	5×10^{-6}	5×10^{-12} 3×10^{-11} 1×10^{-10}	4972	2×10^{-5}	0.01	1.67×10^{-5}	1.63×10^{-5}	2.40	0.82
						4.44×10^{-5}	4.40×10^{-5}	0.90	2.20
						9.55×10^{-5}	9.51×10^{-5}	0.42	4.76
						2.42×10^{-5}	2.37×10^{-5}	2.07	1.19
6	1×10^{-5} 3.33×10^{-6} 5×10^{-6}	2×10^{-11} 6.67×10^{-12} 1×10^{-11}	2500	2×10^{-5}	0.01	2.38×10^{-5}	2.30×10^{-5}	3.36	1.15
						1.78×10^{-5}	1.78×10^{-5}	0.00	0.89
						2.94×10^{-5}	2.80×10^{-5}	4.76	1.40
						1.05×10^{-5}	1.11×10^{-5}	5.71	11.10
1	1×10^{-6}	1×10^{-11}	4972	1×10^{-6} 5×10^{-6} 1×10^{-5} 3×10^{-5} 4×10^{-5} 5×10^{-5} 2×10^{-5}	0.01	1.12×10^{-5}	1.20×10^{-5}	7.14	2.40
						1.28×10^{-5}	1.33×10^{-5}	3.91	1.33
						1.82×10^{-5}	1.86×10^{-5}	2.20	0.62
						2.02×10^{-5}	2.13×10^{-5}	5.45	0.53
6	5×10^{-6}	1×10^{-11}	4972	5×10^{-6} 1×10^{-5} 3×10^{-5} 4×10^{-5} 5×10^{-5}	0.01	1.98×10^{-5}	2.01×10^{-5}	1.52	4.02
						2.06×10^{-5}	2.11×10^{-5}	2.43	2.11
						2.62×10^{-5}	2.53×10^{-5}	3.44	0.84
						2.75×10^{-5}	2.75×10^{-5}	0.00	0.69
1	1×10^{-6}	1×10^{-11}	4972	2×10^{-5}	0.005 0.02 0.03	2.96×10^{-5}	2.98×10^{-5}	0.68	0.60
						1.50×10^{-5}	1.54×10^{-5}	2.67	0.77
						1.59×10^{-5}	1.64×10^{-5}	3.14	0.82
						1.63×10^{-5}	1.67×10^{-5}	2.45	0.84
6	5×10^{-6}	1×10^{-11}	4972	2×10^{-5}	0.005 0.02 0.03	2.33×10^{-5}	2.28×10^{-5}	2.15	1.14
						2.35×10^{-5}	2.36×10^{-5}	0.43	1.18
						2.37×10^{-5}	2.39×10^{-5}	0.84	1.20
						Input			
κ	W	U	G	$\bar{\sigma}$	V	Simulation	Eq. (23)	Error (%)	
(c) Asperity load ratio									
1	1×10^{-6}	1×10^{-11}	4972	2×10^{-5}	0.01	34.65	35.61	0.96	0.80
2						31.86	35.61	3.75	1.08
3						32.53	35.61	3.08	1.22
4						34.13	35.61	1.48	1.28
6	1.5×10^{-7} 5×10^{-6} 2×10^{-5} 1×10^{-6}	1×10^{-11}	4972	2×10^{-5}	0.01	35.77	35.61	0.16	1.31
8						37.65	35.61	2.04	1.31
1						63.12	54.76	8.36	0.97
1						22.42	23.25	0.83	0.67
1	1×10^{-6}	5×10^{-12} 3×10^{-11} 1×10^{-10}	4972	2×10^{-5}	0.01	14.78	15.14	0.36	0.59
						52.91	53.9	0.99	0.60
						7.09	7.79	0.70	1.41
						0.03	0.37	0.34	3.00
1	2×10^{-6} 6.67×10^{-7}	2×10^{-11} 6.67×10^{-12}	2500	2×10^{-5}	0.01	24.34	26.24	1.90	0.80
						41.84	42.08	0.24	0.80

1 × 10 ⁻⁶	2500	1 × 10 ⁻¹¹	47.01	47.6	0.59	0.66
5 × 10 ⁻⁶	7500	1 × 10 ⁻¹¹	25.13	25.75	0.62	0.93
1 × 10 ⁻⁵	4972	2 × 10 ⁻⁵	23.8	23.25	0.55	1.16
1 × 10 ⁻⁴			20.73	18.92	1.81	1.10
5 × 10 ⁻⁶	4972	5 × 10 ⁻¹²	12.62	8.41	4.21	0.92
		3 × 10 ⁻¹¹	45.17	39.47	5.70	0.82
		1 × 10 ⁻¹⁰	1.65	3.51	1.86	2.20
1 × 10 ⁻⁵	2500	2 × 10 ⁻⁵	0	0.15	0.15	4.76
3.33 × 10 ⁻⁶	7500		15.73	16.55	0.82	1.19
5 × 10 ⁻⁶	2500	6.67 × 10 ⁻¹²	29.35	27.92	1.43	1.15
	7500	1 × 10 ⁻¹¹	37.66	34.8	2.86	0.89
1 × 10 ⁻⁶	4972	1 × 10 ⁻¹¹	14.67	14.89	0.22	1.40
			0	0	0.00	11.10
			0.19	0.32	0.13	2.40
			7.81	6.34	1.33	1.33
			54.1	58.2	4.10	0.62
			68.23	74.55	6.32	0.53
			0	0.13	0.13	4.02
			2.25	2.83	0.58	2.11
			44.33	42.44	1.89	0.84
			60.56	56.67	3.89	0.69
			71.01	67.79	3.22	0.60
			25.53	31.58	6.05	0.77
			35.73	39.71	3.98	0.82
			40.4	42.14	1.74	0.84
			18.02	19.99	1.97	1.14
			25	26.65	1.65	1.18
			0.02	28.68	0.13	1.20
			0.03			

The central film thickness equation is

$$H_c = \frac{h_c}{R_x} = (H_c)_{\text{smooth}} \times (1 + 0.025\bar{\sigma}^{1.248} V^{0.119} W^{-0.133} U^{-0.884} G^{-0.977} \kappa^{0.081}) \quad (21)$$

where $(H_c)_{\text{smooth}}$ is obtained from Eq. (19).

The minimum film thickness equation is:

$$H_{\min} = \frac{h_{\min}}{R_x} = (H_{\min})_{\text{smooth}} \times (1 + 0.141\bar{\sigma}^{1.073} V^{0.149} W^{-0.044} U^{-0.828} G^{-0.954} \kappa^{-0.395}) \quad (22)$$

where $(H_{\min})_{\text{smooth}}$ is obtained from Eq. (20).

Finally, the asperity load ratio equation (as percentage) is:

$$L_a = 10W^{-0.083} U^{0.143} G^{0.314} [\ln(1 + \bar{\sigma}^{4.689} V^{0.509} W^{-0.501} U^{-2.90} G^{-2.870})] \quad (23)$$

A comparison between the simulation and curve-fit results for the central film thickness, minimum film thickness, and asperity load ratio are shown in Table 4. The mean error values for central film thickness, minimum film thickness, and asperity load ratio are 2.00%, 2.68%, and 1.98%, respectively, while the maximum error values are 6.73%, 8.00%, and 8.36%, respectively. It should be noted that the error for the film thickness is defined as $100 \times |H_{\text{(simulation)}} - H_{\text{(curve-fit)}}| / H_{\text{(simulation)}}$ while the error for the asperity load ratio is defined as $|L_a(\text{simulation}) - L_a(\text{curve-fit})|$.

It is interesting to note that the asperity load ratio equation (Eq. (23)) is not a function of ellipticity. In fact, the results of the simulations suggest that this parameter is only slightly dependent on the ellipticity (See Table 4(c)). This is due to existence of two opposing factors. On the one hand, increasing the ellipticity increases the film thickness (See Table 4(a) and (b)), which results in less asperity interaction. On the other hand, increasing the ellipticity yields a smaller contact pressure which increases the ratio of the load carried by the asperities. The contribution of these two factors may result in increasing or decreasing the asperity load ratio. However, the results of our simulations show that these opposing factors nearly cancel each other, and therefore the asperity load ratio is not highly influenced by the ellipticity parameter.

It is important to mention that the modified Reynolds equation by Patir and Cheng [16] is valid for the film parameter values $(\Lambda = h_{\min}/\sigma = H_{\min}/\bar{\sigma})$ greater than 0.5. Therefore, the film parameter should be checked after calculating the film thickness from the present formulas to ensure the validity of the results. For the same reason, the regression analysis is only done with the results of simulations satisfying this condition (the film parameter of each case is shown in Table 4). It is also advised that when the obtained asperity load ratio from Eq. (23) is very large (typically more than about 70%), the results should be used with caution, even if the film parameter is within the valid range [33]. This is because the behavior of the problem becomes close to dry contact at these cases. This condition generally occurs at very rough surfaces under low load.

5.2.1. An illustrative example

Consider a deep groove ball bearing (Fig. 7) with the inner raceway radius of $R_i = 50$ mm and outer raceway radius of $R_o = 72.5$ mm. The radii of the balls and the grooves are $R_b = 11$ mm and $R_g = 12$ mm. Therefore, the equivalent contact radii are calculated as $R_x = (1/11 + 1/50)^{-1} = 9$ mm and $R_y = (1/11 - 1/12)^{-1} = 132$ mm at the inner race and $R_x = (1/11 - 1/72.5)^{-1} = 13$ mm and $R_y = (1/11 - 1/12)^{-1} = 132$ mm at the outer race. This gives $D = R_y/R_x = 14.7$ for the inner race and $D = 10.2$ for the outer race contact. Therefore, from Eq. (9), the contact ellipticity is $\kappa = 5.7$ for the inner race and $\kappa = 4.6$ for the outer race.

The shaft (and therefore inner race) rotational speed is 2000 RPM ($\omega=209.4$ rad/s), while the outer race is fixed. It gives the orbital speed of $\omega_m=86$ rad/s and roller rotational speed of $\omega_b=564$ rad/s which yields the rolling speed of $u=6.2$ m/s at both inner and outer race contact. For more information about calculating the speeds, the reader is referred to Ref. [45]. It is also assumed that the bearing is under a radial load where a maximum normal load of $F=500$ N is applied on each ball.

The bearing surfaces (both balls and grooves) are polished with the surface roughness of $\sigma=0.1$ μm which gives the combined roughness of 0.14 μm . The contact material is stainless steel with the effective Young's modulus of $E'=226$ GPa and the Vickers hardness of the surfaces is 600 Kg/mm² (5.89 GPa). Also, the lubricant has the inlet viscosity of $\mu_0=0.03$ Pa.s and pressure-viscosity coefficient of $\alpha=2 \times 10^{-8}$ m²/N.

Considering the above data, Eq. (7) yields $W=2.73 \times 10^{-5}$, $U=9.14 \times 10^{-11}$, and $\sigma=1.56 \times 10^{-5}$ for the inner race contact and $W=1.31 \times 10^{-5}$, $U=6.33 \times 10^{-11}$, and $\sigma=1.08 \times 10^{-5}$ for the outer race contact. Also, $G=4520$ and $V=0.026$ are obtained. Substituting the dimensionless values into Eqs. (21) and (22) yields $H_c=8.87 \times 10^{-5}$ and $H_{\min}=7.47 \times 10^{-5}$ at the inner race, so $h_c=0.80$ μm and $h_{\min}=0.67$ μm . For the outer race, $H_c=7.09 \times 10^{-5}$ and $H_{\min}=5.95 \times 10^{-5}$, so $h_c=0.91$ μm and $h_{\min}=0.77$ μm . The film parameter is obtained as $\Lambda=h_{\min}/\sigma=4.79$ for the inner race and $\Lambda=5.51$ for the outer race. Also, Eq. (23) yields an asperity load ratio of $L_a=0.05\%$ for the inner race and $L_a=0.03\%$ for the outer race which means that the asperity contact is negligible due to high speed and low surface roughness.

Now consider that the shaft rotational speed is one tenth of the previous case (200 RPM). This gives the rolling speed of 0.62 m/s, therefore the dimensionless speed is obtained as $U=9.14 \times 10^{-12}$ at the inner race and $U=6.33 \times 10^{-12}$ at the outer race. In this case, the formulas give $H_c=1.98 \times 10^{-5}$ and $H_{\min}=1.81 \times 10^{-5}$ at the inner race, so $h_c=0.18$ μm and $h_{\min}=0.16$ μm . For the outer

race, $H_c=1.59 \times 10^{-5}$ and $H_{\min}=1.44 \times 10^{-5}$, so $h_c=0.20$ μm and $h_{\min}=0.19$ μm . The film parameter is obtained as $\Lambda=h_{\min}/\sigma=1.16$ for the inner race and $\Lambda=1.33$ for the outer race which shows that mixed lubrication regime prevails. The asperity load ratio is obtained as $L_a=12.09\%$ for the inner race and $L_a=10.33\%$ for the outer race which confirms the presence of asperity contact.

6. Verifications of the results

6.1. Smooth surfaces

6.1.1. Comparison with Hamrock and Dowson's formulas

In this section, the results from the present curve-fit formulas for the smooth surface (Eqs. (19) and (20)) are compared with those by Hamrock and Dowson [3]. Both the central and the minimum film thickness are compared for two different ellipticity values ($\kappa=1$ and $\kappa=6$) for a range of dimensionless speed which are shown in Fig. 8.

As shown, the present curve-fit expressions are in agreement with those by Hamrock and Dowson well [3]. The difference between the results may be due to the fact that a different form of expression is utilized in the current study to enhance the accuracy. Also, more number of nodes as well as a wider domain is used in the current simulations compared to those by Hamrock & Dowson. Moreover, the current regression analyses are performed using a larger number of simulation cases.

6.1.2. Comparison with experimental results

Koye and Winer [46] conducted a set of experiments to measure the minimum film thickness in point-contact EHL. They showed that the curve-fit formula by Hamrock and Dowson [3] followed the experimental results well. In Table 5, their experimental results are compared with those obtained from the present curve-fit expression for the minimum film thickness (Eq. (20)) as well as those by Hamrock and Dowson's formula [3].

As shown, the current results from the film thickness formula (Eq. (20)) follow the experimental results. In fact, the agreement between Eq. (20) and the experiments is better than that between Hamrock and Dowson's formula and the experiments. In Table 5, the average error of Eq. (20) with respect to experimental data is 6.7%, while it is 11.4% for Hamrock & Dowson's formula compared to experiments.

To explore the source of relatively large error of our formula compared to experiments in a few cases of Table 5 (12.07% at case #3, 19.46% at case #9, and 11.88% at case #11), these cases were solved numerically. The results of the simulations showed the associated error of 14.8% at case #3, 20% at case #9, and 1.3% at case #11. This implies that the source of error in case #3 & #9 is

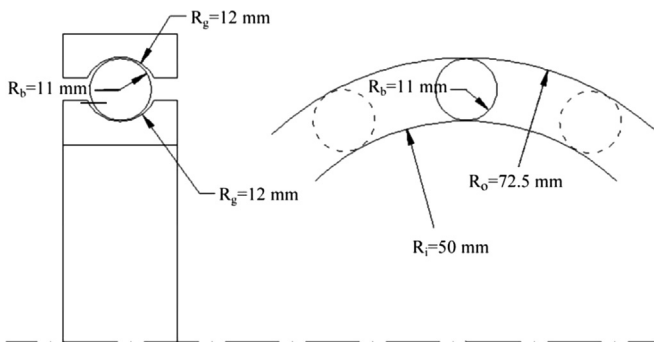


Fig. 7. Ball bearing configuration.

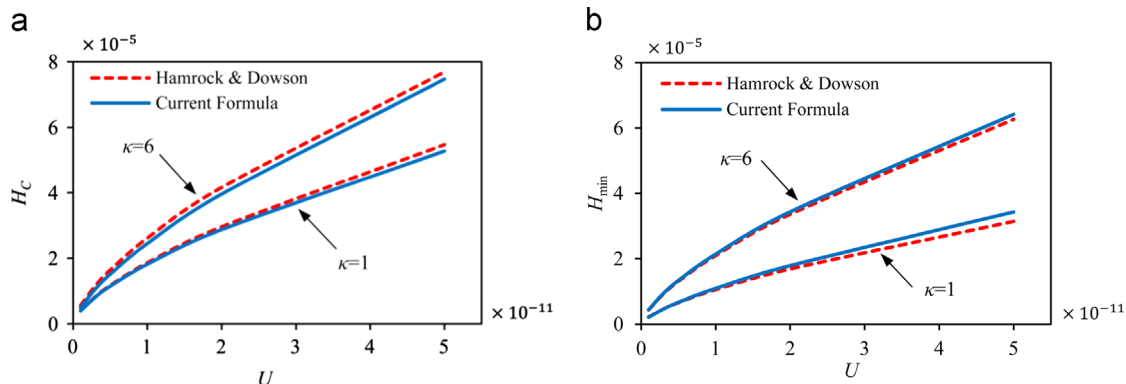


Fig. 8. Comparison between present film thickness formulas for smooth surface and formula by Hamrock and Dowson [3] ($W=1 \times 10^{-6}$ and $G=4972$) (a) central film thickness (Eq. (19)) and (b) minimum film thickness (Eq. (20)).

Table 5

Comparison between experiments [46], Hamrock and Dowson's formula [3], and current results (Eq. (20)).

Case no.	κ	W	U	$H_{\min}$ from experiment [46]	$H_{\min}$ from present expression (Eq. (20))	Error % of Eq. (20) compared to experiments	$H_{\min}$ by Hamrock and Dowson [3]	Error % of Hamrock and Dowson compared to experiments
1	3.7	1.61×10^{-6}	3.53×10^{-11}	6.85×10^{-5}	7.21×10^{-5}	5.33	6.43×10^{-5}	6.16
2	3.7	1.61×10^{-6}	6.33×10^{-11}	1.11×10^{-4}	1.08×10^{-4}	2.92	9.56×10^{-5}	14.01
3	3.7	1.61×10^{-6}	8.80×10^{-11}	1.21×10^{-4}	1.35×10^{-4}	12.07	1.20×10^{-4}	1.05
4	3.7	5.31×10^{-6}	3.53×10^{-11}	7.06×10^{-5}	6.61×10^{-5}	6.44	5.89×10^{-5}	16.54
5	3.7	5.31×10^{-6}	6.33×10^{-11}	1.04×10^{-4}	9.88×10^{-5}	4.70	8.76×10^{-5}	15.48
6	3.7	5.30×10^{-6}	8.80×10^{-11}	1.32×10^{-4}	1.24×10^{-4}	5.80	1.10×10^{-4}	16.73
7	2.4	9.68×10^{-7}	3.48×10^{-11}	6.44×10^{-5}	6.60×10^{-5}	2.54	5.78×10^{-5}	10.21
8	2.4	9.68×10^{-7}	6.25×10^{-11}	9.50×10^{-5}	9.93×10^{-5}	4.54	8.61×10^{-5}	9.36
9	2.4	9.68×10^{-7}	8.69×10^{-11}	1.05×10^{-4}	1.25×10^{-4}	19.46	1.08×10^{-4}	3.00
10	2.4	3.26×10^{-6}	3.48×10^{-11}	6.03×10^{-5}	6.00×10^{-5}	0.50	5.29×10^{-5}	12.25
11	2.4	3.26×10^{-6}	6.25×10^{-11}	1.02×10^{-4}	9.02×10^{-5}	11.88	7.88×10^{-5}	23.05
12	2.4	3.26×10^{-6}	8.69×10^{-11}	1.09×10^{-4}	1.14×10^{-4}	4.44	9.86×10^{-5}	9.30

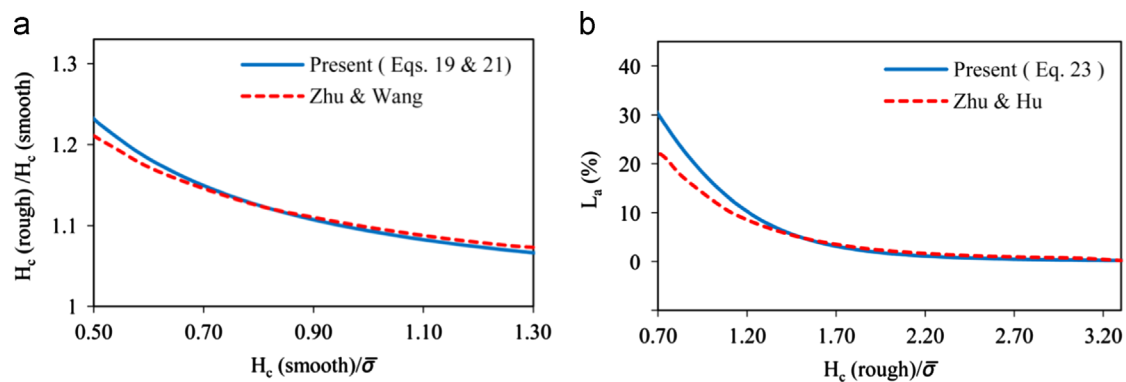


Fig. 9. Verification of present formulas for rough surfaces (a) results from central film thickness formula (Eq. (21)) and simulations by Zhu and Wang [26] ($\kappa=2$, $W=5.3 \times 10^{-5}$ and $G=2830$, $\sigma=4.7 \times 10^{-5}$) and (b) results from asperity load ratio formula (Eq. (23)) and simulations by Zhu and Hu [23] ($\kappa=1$, $W=1.008 \times 10^{-5}$, $G=4000$, $\sigma=2.1 \times 10^{-5}$).

not due to curve-fitting. However, it seems that the error in case #11 is coming from the expression. It should be mentioned that the experimental tests [46] were conducted under the dimensionless material number of $G=10451$ which is larger than the maximum value used in the regression analyses of the current study and those by Hamrock and Dowson. Therefore, the results from the expressions should be taken by caution.

6.2. Rough surfaces

In this section, the results from the present curve-fit formulas for the film thickness and asperity load ratio in rough point-contact EHL are compared with the simulations results by and Zhu and Wang [26] and Zhu and Hu [23] which are both based on deterministic method for characterizing the surface roughness. In deterministic treatment of surface roughness, an actual surface profile (or a randomly-generated surface profile using a given surface roughness value) is directly entered into the deformation equations. As a result, the obtained film thickness profile in such studies has fluctuations and therefore its average value within a specific distance from the contact center is reported as the mean film thickness.

Fig. 9(a) shows the comparison between the results of the present curve-fit formula for central film thickness (Eqs. (19) and (21)) and those obtained in deterministic simulations by Zhu and Wang [26]. The input is $\kappa=2$, maximum Hertzian pressure of 2.277 GPa (equal to $W=5.3 \times 10^{-5}$), $G=2830$ and $\sigma=4.7 \times 10^{-5}$. In the study by Zhu and Wang, the rolling speed is varied and the rough-to-smooth film thickness ratio is plotted against the ratio of the central film thickness (smooth) to the surface roughness. In

their study [26], the rough-to-smooth ratio of the film thickness (vertical axis) is obtained using their simulations while the smooth film thickness in the horizontal axis is obtained by Hamrock and Dowson's formula [3]. In the current study, the results from the present smooth central film thickness formula (Eq. (19)) is used for the horizontal axis while the results from the present rough and smooth central film thickness formulas (Eqs. (19) and (21)) are used for the vertical axis.

As shown in Fig. 9(a), the results from the current formulas are in close agreement with the simulation results by Zhu and Wang. At $H_c(\text{smooth})/\sigma=0.5$ (occurs at $U=2.40 \times 10^{-11}$), their simulation shows the rough-to-smooth film ratio of 1.21 while the present formulas predict it as 1.23. Also, at $H_c(\text{smooth})/\sigma=1.3$ ($U=1 \times 10^{-10}$), both their simulation and current formulas show this ratio is about 1.07.

It should be noted that in Fig. 9(a), the minimum value of horizontal axis is chosen as $H_c(\text{smooth})/\sigma=0.5$, since at this point the film parameter $\Lambda=H_{\min}/\sigma$ (where $H_{\min}$ can be obtained from Eq. (22)) is obtained as $\Lambda=0.51$. This is close to the critical value of 0.5 below which using the modified Reynolds equation by Patir & Cheng [16] is not recommended.

It is worthwhile to mention that the current film thickness formula (Eq. (21)) gives the central value of the film thickness which is the distance between the mean lines of two rough surfaces at the center of the contact (statistical approach). In Fig. 9(a), this value is compared to the mean value of film thickness in the study by Zhu and Wang which is obtained by averaging the film thickness within the half radius from the center of normalized Hertzian contact zone (deterministic approach).

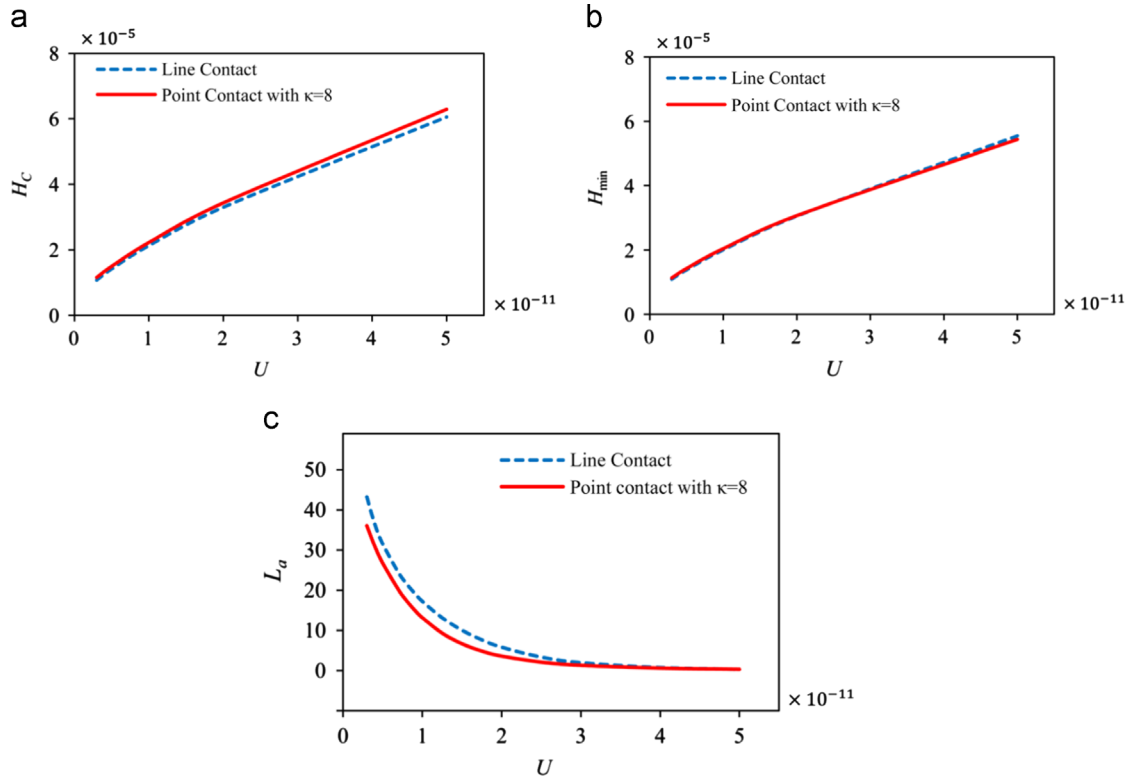


Fig. 10. Comparison between point-contact EHL formulas with $\kappa=8$ and line-contact EHL formulas [33] ($W_{point}=3 \times 10^{-5}$, $W_{line}=9.8 \times 10^{-5}$, $G=4972$, $\sigma=2 \times 10^{-5}$, $V=0.01$) (a) central film thickness (b) minimum film thickness and (c) asperity load ratio.

To verify the obtained asperity load ratio, the results from the current formula (Eq. (23)) are compared to those reported by Zhu and Hu [23]. Fig. 9(b) shows this comparison. The input used in their study is $\kappa=1$, $W=1.008 \times 10^{-5}$, $G=4000$, and $\sigma=2.10 \times 10^{-5}$. Here, the horizontal axis shows the ratio of the average rough film thickness (equivalent to central film thickness in the current study) to the surface roughness.

As shown, the results from the current curve-fit expression (Eq. (23)) are in agreement with the simulation results by Zhu and Hu. However, at lower film thickness values, the present formula predicts a higher asperity load ratio. In Fig. 9(b), the maximum difference between the predicted asperity load ratios is about 8% (at $H_c/\sigma=0.70$ occurs at $U=7.6 \times 10^{-12}$) where their simulation shows $L_a=22\%$ and the present formula (Eq. (23)) predicts $L_a=30\%$. This difference becomes less than 3% after $H_c/\sigma=1$. Note that in Fig. 9(b), the minimum value of horizontal axis is chosen as $H_c(\text{rough})/\sigma=0.70$, because the film parameter ($\Lambda=H_{min}/\sigma$) at this point reaches the critical value of 0.5.

It should be mentioned that since the surface hardness is not reported in neither of the studies by Zhu and Wang [26] and Zhu and Hu [23], a dimensionless hardness of 0.01 (equal to about 20 Rockwell C) is used as input in the current formulas.

The agreement between the results of the present study and the studies by Zhu & Wang [26] and Zhu and Hu [23] is especially interesting because of the inherently different approach employed. While the current formulas are based on the result of simulations with statistical treatment of the surface asperities, the method used in both studies by Zhu & Wang and Zhu & Hu are based on the deterministic method. However, despite this difference, it is shown that both methods predict close results. While the deterministic approach has the advantage of going through boundary lubrication regime (lower lambda ratios), the main advantage of using the formulas reported in this paper is that the film thickness and asperity load ratio can be readily predicted without the need of going through the numerical simulations.

6.3. Comparison with line-contact EHL Results

In the study by Hamrock and Dowson [2], it was demonstrated that at the ellipticity of $\kappa=8$, the film thickness obtained by point-contact EHL solution becomes close to that obtained from line-contact solution. Here, the results from the present film thickness and asperity load ratio formulas for rough point-contact EHL (Eqs. (21)–(23)) at $\kappa=8$ are compared with those presented by the authors for rough line-contact EHL [33].

In order to be able to compare the point contact and line contact results, an equivalent load should be used since the dimensionless load in line contact ($W_{line}=F/BE'R$) contains the contact length (B). Therefore, for a given point-contact problem with $\kappa=8$ and given load, the Hertzian contact length ($2a$) is calculated using Eq. (8) and put as the contact length B . That is:

$$W_{point} = \frac{F}{E'R_x^2}, \quad W_{line} = \frac{F}{BE'R_x} = \frac{W_{point}E'R_x^2}{2aE'R_x} = \frac{W_{point}R_x}{2a} = \frac{W_{point}R_x}{2\kappa b} = \frac{W_{point}}{2\kappa\gamma} \quad (24)$$

where γ is obtained from Eq. (8) using the given load (W_{point}) and ellipticity ($\kappa=8$).

Fig. 10 illustrate the comparison between the results of the current formulas for point-contact EHL (Eqs. (21)–(23)) at ellipticity of $\kappa=8$ and those for line-contact EHL presented in Ref. [33]. The central and the minimum film thickness as well as the asperity load ratio are compared. The input used is fixed at $W_{point}=3 \times 10^{-5}$ (equivalent to $W_{line}=9.8 \times 10^{-5}$), $G=4972$, $\sigma=2 \times 10^{-5}$, and $V=0.01$. The dimensionless speed is varied along the horizontal axis between $U=3 \times 10^{-12}$ (below which the film parameter become smaller than the critical value of 0.5) to $U=5 \times 10^{-11}$ (beyond which there is almost no effect of surface roughness).

As depicted in Fig. 10, at ellipticity of $\kappa=8$, the results obtained from point-contact EHL formulas (Eqs. (21)–(23)) are in good agreement with those obtained from line-contact EHL formulas

presented in Ref. [33]. Therefore, based on the results of this study, it is suggested that for the ellipticity range of $\kappa=1$ (spherical contact) to $\kappa=8$, point-contact formulas (Eqs. (21)–(23)) can be utilized, while for ellipticity values above $\kappa=8$ (equivalent to $D=R_y/R_x > 25.3$), line-contact formulas [33] can be used.

7. Conclusions

In this paper, a model is developed to investigate the effect of surface roughness in elastohydrodynamic lubrication of elliptical contacts. The modified Reynolds, surface deformation and elastoplastic asperity contact equations are simultaneously solved in dimensionless form. The results of an extensive set of simulations are then used to obtain the following curve-fit expressions for the film thickness and asperity load ratio.

Central film thickness

$$H_c = 3.672W^{-0.045\kappa^{0.18}}U^{0.663\kappa^{0.025}}G^{0.502\kappa^{0.064}}(1 - 0.573e^{-0.74\kappa})(1 + 0.025\bar{\sigma}^{1.248}V^{0.119}W^{-0.133}U^{-0.884}G^{-0.977}\kappa^{0.081})$$

Minimum film thickness

$$H_{\min} = 1.637W^{-0.09\kappa^{-0.15}}U^{0.711\kappa^{-0.023}}G^{0.65\kappa^{-0.045}}(1 - 0.974e^{-0.676\kappa})(1 + 0.141\bar{\sigma}^{1.073}V^{0.149}W^{-0.044}U^{-0.828}G^{-0.954}\kappa^{-0.395})$$

Asperity load ratio

$$L_a = 10W^{-0.083}U^{0.143}G^{0.314}[\ln(1 + \bar{\sigma}^{4.689}V^{0.509}W^{-0.501}U^{-2.90}G^{-2.870})]$$

The results from these formulas are shown to be in agreement with the results presented in other publications for both smooth and rough surfaces. It is also shown that at higher ellipticity values, the results of point-contact EHL become close to those of line-contact EHL.

The present formulas have the advantage of being easy-to-use without the need of performing extensive numerical simulations. Using these formulas, one can readily predict the film thickness and therefore the film parameter for various operating conditions. This will determine if the machine elements are working within the safe conditions or they are prone to reach the boundary lubrication

regime and ultimate failure. The asperity load ratio equation is also a key factor for calculating the friction and the wear.

Appendix A. Finite difference scheme

The Reynolds equation (Eq. (11)) is discretized as

$$A_{ij}P_{hi+1,j} + B_{ij}P_{hi,j-1} + C_{ij}P_{hi-1,j} + D_{ij}P_{hi+1,j+1} - E_{ij}P_{hi,j} = F_{ij}$$

where

$$A_{ij} = 3\Phi_{i+1,j}\eta_{i+1,j}H_{i+1,j}^3 + \Phi_{i-1,j}\eta_{i-1,j}H_{i-1,j}^3$$

$$B_{ij} = \left(\frac{\Delta X}{\kappa\Delta Y}\right)^2 [\Phi_{ij+1}\eta_{ij+1}H_{ij+1}^3 + 3\Phi_{ij-1}\eta_{ij-1}H_{ij-1}^3]$$

$$C_{ij} = \Phi_{xi+1,j}\eta_{i+1,j}H_{i+1,j}^3 + 3\Phi_{i-1,j}\eta_{i-1,j}H_{i-1,j}^3$$

$$D_{ij} = \left(\frac{\Delta X}{\kappa\Delta Y}\right)^2 [3\Phi_{ij+1}\eta_{ij+1}H_{ij+1}^3 + \Phi_{ij-1}\eta_{ij-1}H_{ij-1}^3]$$

$$E_{ij} = 4(\Phi_{i+1,j}\eta_{i+1,j}H_{i+1,j}^3 + \Phi_{i-1,j}\eta_{i-1,j}H_{i-1,j}^3)$$

$$+ 4\left(\left(\frac{\Delta X}{\kappa\Delta Y}\right)^2 [\Phi_{ij+1}\eta_{ij+1}H_{ij+1}^3 + \Phi_{ij-1}\eta_{ij-1}H_{ij-1}^3]\right)$$

$$F_{ij} = 2K\Delta X(3\bar{\rho}_{ij}\eta_{ij}H_{Tij} - 4\bar{\rho}_{i-1,j}\eta_{ij}H_{Ti-1,j} + \bar{\rho}_{i-2,j}\eta_{ij}H_{Ti-2,j})$$

in which

$$\eta = \frac{\bar{\rho}}{\bar{\mu}}, \quad \Phi = \Phi_x = \Phi_y, \quad K = \frac{8\pi\kappa\gamma^3 U}{W}$$

Note that a combination of central and forward/backward scheme is utilized for the left-hand side, while a second order backward scheme is used for the right-hand side.

Appendix B. Numerical solution procedure

- The over-relaxation factor for S.O.R is generally chosen as 1.6.
- The under-relaxation factor for pressure is generally chosen at 0.01 and gradually decreased during the procedure to stabilize the results.
- Parameter H_{00} is under-relaxed by the load balance once in every 10 loops. Its under-relaxation factor is generally chosen as 0.01 (Fig. B1)

Appendix C. Effect of inlet size

Smooth surface, low load ($W=1 \times 10^{-6}$, $U=1 \times 10^{-11}$, $G=4000$, $\kappa=3$)

Parameter*	$X_{\min} = -2$	$X_{\min} = -3$	$X_{\min} = -4$	$X_{\min} = -5$
H_c (Error)	1.879×10^{-5} (–)	2.016×10^{-5} (7.29%)	2.068×10^{-5} (2.58%)	2.074×10^{-5} (0.29%)
$H_{\min}$ (Error)	1.599×10^{-5} (–)	1.679×10^{-5} (5.00%)	1.698×10^{-5} (1.13%)	1.704×10^{-5} (0.35%)

Smooth surface, high load ($W=1 \times 10^{-5}$, $U=1 \times 10^{-11}$, $G=4000$, $\kappa=3$)

Parameter*	$X_{\min} = -2$	$X_{\min} = -3$	$X_{\min} = -4$	$X_{\min} = -5$
H_c (Error)	1.559×10^{-5} (–)	1.583×10^{-5} (1.54%)	1.593×10^{-5} (0.61%)	1.598×10^{-5} (0.31%)
$H_{\min}$ (Error)	1.451×10^{-5} (–)	1.441×10^{-5} (0.69%)	1.447×10^{-5} (0.42%)	1.453×10^{-5} (0.41%)

Rough surface, low load ($\bar{\sigma}=2 \times 10^{-5}$, $W=1 \times 10^{-6}$, $U=1 \times 10^{-11}$, $G=4000$, $\kappa=3$)

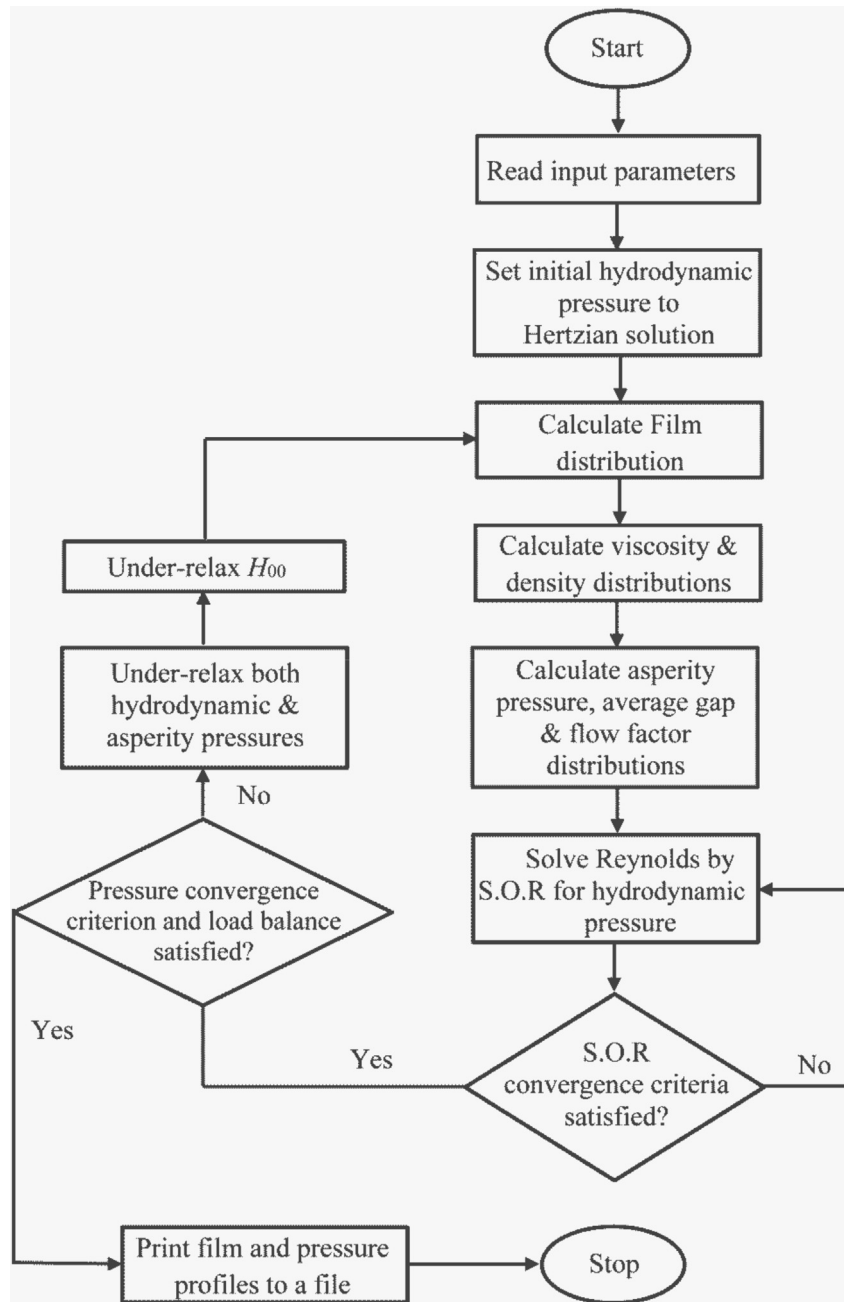


Fig. B1.

Parameter*	$X_{\min} = -2$	$X_{\min} = -3$	$X_{\min} = -4$	$X_{\min} = -5$
H_c (Error)	2.297×10^{-5} (–)	2.443×10^{-5} (6.35%)	2.478×10^{-5} (1.43%)	2.488×10^{-5} (0.40%)
$H_{\min}$ (Error)	2.141×10^{-5} (–)	2.242×10^{-5} (4.72%)	2.267×10^{-5} (1.11%)	2.274×10^{-5} (0.31%)
L_a (Error)	42.148 (–)	37.833 (10.24%)	36.706 (2.98%)	36.486 (0.60%)

Rough surface, high load ($\bar{\sigma} = 2 \times 10^{-5}$, $W = 1 \times 10^{-5}$, $U = 1 \times 10^{-11}$, $G = 4000$, $\kappa = 3$)

Parameter*	$X_{\min} = -2$	$X_{\min} = -3$	$X_{\min} = -4$	$X_{\min} = -5$
H_c (Error)	2.042×10^{-5} (–)	2.057×10^{-5} (0.73%)	2.081×10^{-5} (1.17%)	2.085×10^{-5} (0.19%)
$H_{\min}$ (Error)	1.956×10^{-5} (–)	1.967×10^{-5} (0.56%)	1.976×10^{-5} (0.46%)	1.983×10^{-5} (0.35%)
L_a (Error)	22.778 (–)	22.390 (1.70%)	22.324 (0.29%)	22.3397 (0.07%)

*Errors are calculated with respect to previous inlet size: $(Error = 100 \times |X_i - X_{i-1}|/X_{i-1})$

Appendix D. Mesh refinement study

Mesh1: 120 × 20 nodes (20 nodes per unit along X and 10 nodes per unit along Y)

Mesh2: 150 × 30 nodes (25 nodes per unit along X and 15 nodes per unit along Y)

Mesh3: 180 × 40 nodes (30 nodes per unit along X and 20 nodes per unit along Y)

Mesh4: 210 × 50 nodes (35 nodes per unit along X and 25 nodes per unit along Y)

Smooth surface, low load ($W=1 \times 10^{-6}$, $U=1 \times 10^{-11}$, $G=4000$, $\kappa=3$)

Parameter*	Mesh 1	Mesh 2	Mesh 3	Mesh 4
H_c (Error)	2.047×10^{-5} (–)	2.058×10^{-5} (0.53%)	2.068×10^{-5} (0.48%)	2.077×10^{-5} (0.43%)
$H_{\min}$ (Error)	1.710×10^{-5} (–)	1.703×10^{-5} (0.41%)	1.698×10^{-5} (0.29%)	1.695×10^{-5} (0.18%)

Smooth surface, high load ($W=1 \times 10^{-5}$, $U=1 \times 10^{-11}$, $G=4000$, $\kappa=3$)

Parameter*	Mesh 1	Mesh 2	Mesh 3	Mesh 4
H_c (Error)	1.643×10^{-5} (–)	1.607×10^{-5} (1.59%)	1.593×10^{-5} (0.87%)	1.581×10^{-5} (0.75%)
$H_{\min}$ (Error)	1.481×10^{-5} (–)	1.459×10^{-5} (1.49%)	1.447×10^{-5} (0.82%)	1.438×10^{-5} (0.62%)

Rough surface, low load ($\bar{\sigma}=2 \times 10^{-5}$, $W=1 \times 10^{-6}$, $U=1 \times 10^{-11}$, $G=4000$, $\kappa=3$)

Parameter*	Mesh 1	Mesh 2	Mesh 3	Mesh 4
H_c (Error)	2.474×10^{-5} (–)	2.477×10^{-5} (0.12%)	2.478×10^{-5} (0.04%)	2.479×10^{-5} (0.04%)
$H_{\min}$ (Error)	2.261×10^{-5} (–)	2.265×10^{-5} (0.18%)	2.267×10^{-5} (0.09%)	2.268×10^{-5} (0.04%)
L_a (Error)	36.785 (–)	36.733 (0.14%)	36.706 (0.07%)	36.693 (0.04%)

Rough surface, high load ($\bar{\sigma}=2 \times 10^{-5}$, $W=1 \times 10^{-5}$, $U=1 \times 10^{-11}$, $G=4000$, $\kappa=3$)

Parameter*	Mesh 1	Mesh 2	Mesh 3	Mesh 4
H_c (Error)	2.051×10^{-5} (–)	2.079×10^{-5} (1.37)	2.081×10^{-5} (0.09%)	2.083×10^{-5} (0.10%)
$H_{\min}$ (Error)	1.956×10^{-5} (–)	1.982×10^{-5} (1.31)	1.976×10^{-5} (0.30%)	1.971×10^{-5} (0.25%)
L_a (Error)	22.429 (–)	22.328 (0.45%)	22.324 (0.02%)	22.331 (0.03%)

*Errors are calculated with respect to previous mesh size: ($Error = 100 \times |X_i - X_{i-1}| / X_{i-1}$)

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